

Receptor Pharmacology - PANCE Summary

Adrenergic, Cholinergic, Histamine, Serotonin, Dopamine, GABA, Opioid, NMDA, AT1, Mineralocorticoid, Leukotriene

PANCE relevance: Pharmacology is tested under *Pharmaceutical Therapeutics (15%)* and *Applying Foundational Scientific Concepts (8%)* - mechanism of action, indications, and ADRs. Drugs listed are PANCE-relevant agents commonly prescribed clinically.

Adrenergic (Sympathetic) Receptors

Receptor	Location & Effect When Activated	Agonists (drug → clinical use)	Antagonists (drug → clinical use)	Key Pearls / ADRs
$\alpha 1$ Gq (IP3/DAG)	Vascular smooth muscle, prostate / bladder neck, iris dilator (radial muscle) → Vasoconstriction, mydriasis, ↑ bladder neck tone (urinary retention)	Phenylephrine → pressor, nasal decongestant Midodrine → orthostatic hypotension Oxymetazoline → intranasal decongestant Norepinephrine → septic shock	Non-selective: prazosin, doxazosin, terazosin → HTN, BPH, PTSD nightmares (prazosin) $\alpha 1A$ -selective: tamsulosin, silodosin, alfuzosin → BPH <i>Mixed $\alpha 1/\beta$: carvedilol, labetalol</i> (see $\beta 2$ row)	First-dose orthostatic hypotension / syncope Reflex tachycardia Tamsulosin: intraoperative floppy iris syndrome at cataract surgery
$\alpha 2$ Gi	Presynaptic CNS sympathetic neurons; pancreatic β -cells → ↓ sympathetic outflow, ↓ insulin release	Clonidine → HTN (resistant), ADHD, opioid withdrawal, hot flashes Tizanidine → muscle spasticity Dexmedetomidine → ICU sedation Brimonidine → glaucoma (topical) Methylidopa → HTN in pregnancy	Yohimbine (rarely clinical)	Clonidine: rebound HTN if abruptly stopped ; sedation; dry mouth Methylidopa: + Coombs hemolytic anemia
$\beta 1$ Gs (cAMP)	Heart (SA / AV / myocardium); kidney JG cells → ↑ HR, ↑ contractility, ↑ AV conduction, ↑ renin release	Dobutamine → cardiogenic shock, decompensated HF Isoproterenol → bradycardia / AV block (rare)	Cardioselective ($\beta 1 > \beta 2$): metoprolol, atenolol, bisoprolol, esmolol, nebivolol → HTN, HFrEF, post-MI, AFib rate control, SVT, thyroid storm, migraine prophylaxis	Cardioselective preferred in asthma/COPD & diabetics Mask hypoglycemia symptoms (tachycardia) - sweating preserved Avoid abrupt withdrawal (rebound) Contraindications: severe bradycardia, 2nd/3rd degree AV block, decompensated HF
$\beta 2$ Gs	Bronchi, uterus, vascular & skeletal muscle, liver → Bronchodilation, uterine relaxation, vasodilation, ↑ glycogenolysis, ↑ insulin	SABA: albuterol, levalbuterol → asthma rescue, acute COPD LABA: salmeterol, formoterol → asthma maintenance (only with ICS), COPD Terbutaline → tocolysis (preterm labor)	Non-selective β-blockers (pure β): propranolol, nadolol, timolol, sotalol → Propranolol: essential tremor, performance anxiety, thyrotoxicosis, portal HTN, migraine prophylaxis → Timolol: glaucoma (topical) → Sotalol: also class III antiarrhythmic (K ⁺ blocker) for AFib Mixed $\alpha 1$ + non-selective β-antagonists: carvedilol (PO, HFrEF + HTN), labetalol (IV/PO, HTN emergency, HTN in pregnancy/preeclampsia) → $\alpha 1$ blockade adds vasodilation: lowers SVR rapidly (helpful in HTN emergency & pregnancy) and reduces afterload (HF benefit)	$\beta 2$ agonist ADRs: tremor, tachycardia, hypokalemia (K ⁺ shifts intracellularly), hyperglycemia LABA monotherapy → ↑ asthma death (black box) Avoid non-selective β -blockers in asthma Carvedilol/labetalol: orthostatic hypotension from $\alpha 1$ component (start low, titrate up)
$\beta 3$ Gs	Bladder detrusor, brown adipose → Detrusor relaxation (↑ bladder capacity), lipolysis	Mirabegron → overactive bladder (alternative to antimuscarinics; less dry mouth)	-	Mirabegron ADR: HTN , nasopharyngitis, UTI Avoid in uncontrolled HTN

Mixed Sympathomimetics & Dopaminergics

Drug	Receptors	Clinical Use	Pearls
Epinephrine	$\alpha 1, \alpha 2, \beta 1, \beta 2$ (all)	Anaphylaxis (1st line, IM); cardiac arrest; severe asthma; local anesthetic vasoconstriction	Anaphylaxis dose: 0.3 mg IM (adult), 0.15 mg IM (peds <30 kg)

Drug	Receptors	Clinical Use	Pearls
Norepinephrine	α_1 , α_2 , β_1 (minimal β_2)	Septic shock - 1st line vasopressor	More vasoconstriction than epi; less β_2 activity
Dopamine	Dose-dependent: low D1 $\rightarrow$ mid β_1 $\rightarrow$ high α_1	Shock (less common now), bradycardia	No longer 1st line for septic shock (NE preferred)
Pseudoephedrine / Ephedrine	Indirect: $\uparrow$ NE release (α & β)	Nasal decongestant (pseudo); pressor (ephedrine)	Avoid in HTN; pseudoephedrine restricted (meth precursor)
Cocaine / Amphetamines	Indirect: block reuptake / $\uparrow$ release of NE/DA	Toxidrome: HTN, tachycardia, mydriasis, hyperthermia	Avoid β-blockers in cocaine toxicity (unopposed $\alpha \rightarrow$ vasospasm); use benzos

Cholinergic (Parasympathetic) Receptors

Mnemonic for muscarinic effects (parasympathetic): **SLUDGE** - Salivation, Lacrimation, Urination, Defecation, GI cramping, Emesis. Or **DUMBBELS** - Diaphoresis, Urination, Miosis, Bronchoconstriction, Bradycardia, Emesis, Lacrimation, Salivation.

Receptor	Location & Effect When Activated	Agonists / Cholinomimetics	Antagonists (Anticholinergics)	Key Pearls / ADRs
Muscarinic (M1, M2, M3) M1/M3: Gq M2: Gi	M1: CNS (cognition), gastric parietal cells (acid) M2: Heart → ↓ HR, ↓ AV conduction M3: Smooth muscle & glands → bronchoconstriction, GI/GU motility & secretions, miosis (sphincter), accommodation (ciliary)	Direct: Bethanechol → postop urinary retention, neurogenic bladder Pilocarpine → closed-angle glaucoma; xerostomia (Sjögren, post-radiation) Cevimeline → Sjögren xerostomia Indirect (AChE inhibitors): see next section	Atropine → symptomatic bradycardia, organophosphate poisoning, premed for secretions Ipratropium / Tiotropium → COPD, asthma Oxybutynin, Tolterodine, Solifenacin, Darifenacin, Trospium → OAB Scopolamine → motion sickness (patch) Glycopyrrolate → perioperative secretions; hyperhidrosis Benztropine, Trihexyphenidyl → Parkinson tremor, EPS from antipsychotics Dicyclomine, Hyoscyamine → IBS cramping	Anticholinergic toxidrome: 'Hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter, full as a flask' = hyperthermia, dry skin/mouth, flushing, mydriasis & blurred vision, delirium, urinary retention Avoid in: closed-angle glaucoma, BPH, dementia (worsens cognition) Reverse with physostigmine (crosses BBB) Beers list: anticholinergics in elderly
Nicotinic Ligand-gated cation channel Nm (NMJ) Nn (ganglia, adrenal medulla)	Nm: Skeletal muscle NMJ → contraction Nn: Autonomic ganglia (both sympathetic and parasympathetic), adrenal medulla → epi/NE release	Nicotine (smoking, patch/gum) Varenicline → smoking cessation (partial agonist at $\alpha 4\beta 2$ nAChR) Succinylcholine (depolarizing) → rapid sequence intubation	Non-depolarizing NMJ blockers: rocuronium, vecuronium, cisatracurium, atracurium → intubation, surgical paralysis Reverse with neostigmine + glycopyrrolate or sugammadex (roc/vec) Ganglionic blockers: rare clinically	Succinylcholine ADRs: hyperkalemia (avoid in burns / crush / denervation), malignant hyperthermia (treat with dantrolene), bradycardia in peds Varenicline: vivid dreams, neuropsychiatric (boxed warning removed 2016)

Cholinesterase Inhibitors (Indirect Cholinomimetics)

Drug	Use	Pearls / ADRs
Donepezil, Rivastigmine, Galantamine	Alzheimer disease (mild-moderate; rivastigmine also for Parkinson dementia)	ADRs: nausea, diarrhea, bradycardia, syncope, weight loss; modest symptomatic benefit only
Neostigmine, Pyridostigmine	Myasthenia gravis (pyridostigmine = chronic outpatient); reversal of non-depolarizing NMJ blockade (neostigmine + glycopyrrolate)	Do not cross BBB. Cholinergic crisis if overdose → weakness mimicking MG flare
Physostigmine	Reversal of anticholinergic (antimuscarinic) toxicity	Crosses BBB (treats CNS delirium); contraindicated in TCA overdose (seizures, asystole)
Edrophonium	Historical 'Tensilon test' for myasthenia gravis dx (largely replaced by anti-AChR antibody testing)	Very short-acting
Organophosphates (insecticides, nerve agents)	Toxicity - irreversible AChE inhibition	SLUDGE/DUMBBELS + muscle fasciculations + paralysis. Treat with atropine (muscarinic effects) + pralidoxime/2-PAM (regenerates AChE) - give pralidoxime early before 'aging'

Sympathetic vs Parasympathetic Organ Effects

Organ	Sympathetic Effect (Receptor)	Parasympathetic Effect (Receptor)
Heart - SA/AV/contractility	↑ HR, ↑ conduction, ↑ contractility ($\beta 1$)	↓ HR, ↓ AV conduction (M2)
Vascular smooth muscle	Vasoconstriction ($\alpha 1$) / vasodilation in skeletal muscle ($\beta 2$)	No direct innervation (endothelial NO)
Bronchi	Bronchodilation ($\beta 2$)	Bronchoconstriction, ↑ secretions (M3)
GI tract	↓ motility & secretions ($\alpha 2$, $\beta 2$)	↑ motility & secretions (M3)
Bladder - detrusor / sphincter	Detrusor relaxation ($\beta 3$) / sphincter contraction ($\alpha 1$) → retention	Detrusor contraction (M3) / sphincter relaxation → voiding

Organ	Sympathetic Effect (Receptor)	Parasympathetic Effect (Receptor)
Eye - pupil / ciliary	Mydriasis (α 1, radial muscle)	Miosis (M3, sphincter); accommodation for near (M3, ciliary)
Pancreas (insulin)	↓ insulin (α 2)	↑ insulin (M3)
Kidney (renin)	↑ renin (β 1)	-

Other PANCE-Relevant Receptors

Histamine, Serotonin, Dopamine, GABA, Opioid, NMDA, AT1, Mineralocorticoid, Leukotriene

Receptor	Location & Effect	Agonists (drug → clinical use)	Antagonists (drug → clinical use)	Key Pearls / ADRs
HISTAMINE				
H1 Gq	Bronchi, vascular smooth muscle, sensory nerves → Bronchoconstriction, vasodilation, ↑ vascular permeability, pruritus, pain	Endogenous histamine (allergy, mast cells)	1st-gen (sedating, anticholinergic): diphenhydramine, hydroxyzine, promethazine, chlorpheniramine, doxylamine, meclizine → allergy, urticaria, motion sickness (meclizine, scopolamine), insomnia (diphenhydramine), N/V (promethazine, hydroxyzine) 2nd-gen (non-sedating): loratadine, cetirizine, fexofenadine, desloratadine, levocetirizine → allergic rhinitis, urticaria	1st-gen ADRs: anticholinergic toxidrome, sedation, paradoxical excitation in peds Avoid 1st-gen in elderly (Beers list) Promethazine: extravasation tissue necrosis (avoid IV; black box)
H2 Gs	Gastric parietal cells → ↑ Gastric acid secretion	-	Famotidine, Cimetidine, Nizatidine → GERD, PUD, stress ulcer prophylaxis, Zollinger-Ellison (less potent than PPIs)	Cimetidine: CYP450 inhibitor (warfarin, phenytoin, theophylline interactions); antiandrogen effects (gynecomastia, ED, ↑ prolactin) Famotidine preferred - no CYP / antiandrogen issues Tachyphylaxis with chronic use; PPIs more effective for healing PUD
SEROTONIN (5-HT)				
5-HT1B/1D	Cranial vasculature, trigeminal nerve → Vasoconstriction, ↓ CGRP release	Triptans - sumatriptan, rizatriptan, eletriptan, zolmitriptan → acute migraine	-	Contraindicated in CAD, uncontrolled HTN, prior stroke, hemiplegic migraine, pregnancy Serotonin syndrome risk if combined with SSRIs/SNRIs/MAOIs/tramadol
5-HT1A Gi	CNS (anxiety, mood)	Buspirone - partial agonist → generalized anxiety disorder Aripiprazole - partial agonist (also D2 partial agonist)	-	Buspirone: non-sedating, no dependence, no withdrawal; takes 2-4 weeks for effect; no PRN use
5-HT2A Gq	CNS, platelets	LSD, mescaline (recreational)	Atypical antipsychotics: risperidone, olanzapine, quetiapine, ziprasidone, aripiprazole, paliperidone, lurasidone, clozapine Mirtazapine (also α2 antagonist; antidepressant + appetite/sleep) Trazodone (also SARI; insomnia & depression)	Atypicals → lower EPS than typicals but ↑ metabolic syndrome (weight, T2DM, lipids) Clozapine: agranulocytosis (REMS monitoring), seizures, myocarditis - reserved for refractory schizophrenia Trazodone: priapism
5-HT3 Ligand-gated cation	CNS (chemoreceptor trigger zone), GI vagal afferents → Emesis	-	Ondansetron, Granisetron, Dolasetron, Palonosetron → chemotherapy-induced N/V, post-op N/V, hyperemesis gravidarum Mirtazapine (also blocks 5-HT3 - additional antiemetic effect)	QT prolongation with ondansetron (esp. IV); avoid with other QT-prolonging drugs Headache, constipation
5-HT4 Gs	Enteric neurons → ↑ GI motility	Metoclopramide (also D2 antagonist) → gastroparesis, refractory N/V Prucalopride → chronic constipation	-	Metoclopramide: tardive dyskinesia (black box) ; EPS; hyperprolactinemia Avoid in bowel obstruction, Parkinson disease

Receptor	Location & Effect	Agonists (drug → clinical use)	Antagonists (drug → clinical use)	Key Pearls / ADRs
SSRI / SNRI (reuptake inhibitors)	Presynaptic 5-HT (and NE for SNRI) reuptake transporters	SSRIs: fluoxetine, sertraline, citalopram, escitalopram, paroxetine, fluvoxamine SNRIs: venlafaxine, duloxetine, desvenlafaxine, milnacipran → depression, anxiety, OCD, PTSD, neuropathic pain (duloxetine), fibromyalgia	-	Sexual dysfunction, GI upset, ↑ suicidality in young (black box), serotonin syndrome (esp. with MAOIs/triptans/tramadol/linezolid - clonus, hyperthermia, autonomic instability) Discontinuation syndrome (esp. paroxetine, venlafaxine)
DOPAMINE				
D1 Gs	Renal/mesenteric vasculature, direct pathway striatum	Fenoldopam → hypertensive emergency (rare) Low-dose dopamine	-	Generally less PANCE-relevant than D2
D2 Gi	Mesolimbic, nigrostriatal, tuberoinfundibular pathways → Modulates psychosis, movement, prolactin (tonic inhibition)	Bromocriptine, cabergoline - prolactinoma, suppression of lactation Pramipexole, ropinirole - Parkinson disease, restless legs syndrome Levodopa/carbidopa - Parkinson disease (precursor; not a direct receptor agonist) Apomorphine - PD off-episodes Aripiprazole, brexpiprazole, cariprazine - D2 partial agonists (atypical antipsychotics)	Typicals: haloperidol, fluphenazine, chlorpromazine → schizophrenia, acute psychosis, Tourette Atypicals: see 5-HT _{2A} row Antiemetics: metoclopramide, prochlorperazine, promethazine Tetrabenazine, deutetrabenazine (VMAT2 inhibitors) → Huntington chorea, tardive dyskinesia	Antipsychotic ADRs: EPS (acute dystonia, akathisia, parkinsonism, tardive dyskinesia), neuroleptic malignant syndrome (fever, rigidity, autonomic instability, ↑ CK; treat with dantrolene/bromocriptine), hyperprolactinemia (galactorrhea, gynecomastia, amenorrhea), QT prolongation DA agonists for Parkinson: impulse-control disorders (gambling), N/V, orthostasis, hallucinations
GABA				
GABA-A Ligand-gated Cl ⁻ channel	CNS, widespread → Hyperpolarization (inhibitory)	Benzodiazepines (↑ <i>frequency</i> of Cl ⁻ channel opening): alprazolam, lorazepam, diazepam, midazolam, clonazepam, temazepam, oxazepam, chlordiazepoxide → anxiety, panic, alcohol withdrawal, status epilepticus, insomnia, procedural sedation Z-drugs (α1-selective): zolpidem, zaleplon, eszopiclone → insomnia Barbiturates (↑ <i>duration</i> of opening): phenobarbital → status epilepticus, neonatal sz Propofol, etomidate, ketamine adjunct, alcohol	Flumazenil → benzodiazepine reversal (use cautiously - precipitates seizures in chronic users / mixed overdoses)	Benzo ADRs: sedation, falls in elderly, dependence, withdrawal (seizures), respiratory depression with opioids/alcohol Lorazepam, oxazepam, temazepam ("LOT"): no active metabolites - prefer in liver disease & elderly Barbiturates: narrow therapeutic index, induce CYP
GABA-B Gi	CNS, spinal cord → Inhibitory (presynaptic)	Baclofen → spasticity (MS, spinal cord injury, cerebral palsy); intrathecal pump for severe spasticity GHB (recreational/Xyrem for narcolepsy)	-	Avoid abrupt baclofen withdrawal → seizures, hallucinations, hyperthermia (esp. intrathecal)
OPIOID				
μ (mu) Gi (also κ, δ)	CNS (analgesia, euphoria, respiratory depression), GI → Analgesia, sedation, respiratory depression, miosis, constipation, euphoria	Full agonists: morphine, oxycodone, hydrocodone, hydromorphone, fentanyl, methadone, codeine, meperidine, tramadol Partial agonist: buprenorphine (Suboxone = bup + naloxone) → OUD, chronic pain → analgesia, OUD maintenance, cough suppression (codeine), diarrhea (loperamide, diphenoxylate - peripheral)	Naloxone (short-acting, IV/IM/IN) → opioid overdose reversal Naltrexone (long-acting PO/IM Vivitrol) → OUD & AUD maintenance Methylnaltrexone, naloxegol → opioid-induced constipation (peripheral, no analgesia loss)	Triad of overdose: respiratory depression + miosis + ↓ LOC Tolerance to all effects EXCEPT miosis & constipation Methadone: QT prolongation; long half-life Tramadol/Meperidine: seizure risk & serotonin syndrome (also SNRI activity) Codeine: avoid in children (CYP2D6 ultra-rapid metabolizers) Naloxone in chronic opioid users → precipitates withdrawal
NMDA / GLUTAMATE				

Receptor	Location & Effect	Agonists (drug → clinical use)	Antagonists (drug → clinical use)	Key Pearls / ADRs
NMDA Ligand-gated cation	CNS (excitatory) → Excitotoxicity in stroke / neurodegeneration; involved in memory (LTP)	Glutamate (endogenous)	Memantine → moderate-severe Alzheimer disease Ketamine → dissociative anesthesia, refractory depression (esketamine nasal spray), procedural sedation Dextromethorphan → cough; abuse potential PCP (recreational)	Memantine: well-tolerated; can combine with AChE inhibitor Ketamine : emergence reactions (hallucinations - mitigate with benzo); ↑ ICP/IOP; preserves airway reflexes; bronchodilator (useful in asthma) Alcohol withdrawal: NMDA upregulation contributes to seizures/delirium tremens
OTHER MAJOR RECEPTOR ANTAGONIST CLASSES				
AT1 (angiotensin) Gq	Vascular smooth muscle, adrenal cortex, kidney → Vasoconstriction, ↑ aldosterone, Na⁺/H₂O retention	Angiotensin II (endogenous)	ARBs : losartan, valsartan, candesartan, olmesartan, telmisartan, irbesartan, azilsartan → HTN, HFrEF, CKD/diabetic nephropathy, post-MI ACE inhibitors (block AngII formation, not receptor): lisinopril, enalapril, ramipril, benazepril → same indications	Both classes: hyperkalemia, AKI (esp. with bilateral renal artery stenosis), teratogenic (avoid in pregnancy) ACEi: dry cough (bradykinin), angioedema (higher in Black patients) ARBs: less cough/angioedema - alternative to ACEi Do NOT combine ACEi + ARB
Mineralocorticoid (aldosterone)	Renal collecting duct, heart, vascular smooth muscle → Na⁺/H₂O retention, K⁺ excretion, cardiac fibrosis	Aldosterone (endogenous; pathologic in primary hyperaldosteronism / Conn)	Spironolactone, Eplerenone → HFrEF, resistant HTN, primary hyperaldosteronism, ascites/cirrhosis, hirsutism & acne (spironolactone), PCOS	Both: hyperkalemia Spironolactone: gynecomastia, menstrual irregularity, ↓ libido (eplerenone has fewer endocrine effects) Avoid combining with ACEi/ARB or K ⁺ supplements without close monitoring
Leukotriene (CysLT1)	Bronchi, eosinophils → Bronchoconstriction, mucus, inflammation	Endogenous leukotrienes (esp. LTC ₄ , D ₄ , E ₄)	Montelukast, Zafirlukast (LTRA) → asthma maintenance, allergic rhinitis, exercise-induced bronchospasm Zileuton (5-LOX inhibitor) → asthma	Montelukast: black box for neuropsychiatric effects (depression, suicidality, sleep disturbance) Zileuton: hepatotoxicity - monitor LFTs Especially useful for aspirin-exacerbated respiratory disease (Samter triad)